# Rahul Kaushal

Bengaluru, India

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## Summary

Backend and platform software engineer with 3+ years at Honeywell building Python and Go backend services, data pipelines and edge platform software (REST/WebSocket APIs, event-driven processing, Kubernetes, Ansible, CI/CD) on Azure, Databricks and on-prem appliances. Also worked on MLOps platforms and LLM/RAG applications. Co-inventor on a US patent.

## Technical Skills

**Languages:** Python, Go, SQL, Bash
**Backend & APIs:** FastAPI, REST APIs, WebSocket, asyncio, microservices, event-driven architecture, multi-tenant systems, Pydantic, SQLAlchemy, automated testing (pytest, Behave BDD, Locust)
**Cloud & DevOps:** Kubernetes (MicroK8s, AKS), Docker, Ansible, CI/CD (GitHub Actions), Linux, Azure (Event Hub, IoT Hub, Blob Storage), infrastructure automation, high availability (Pacemaker/Corosync), Ceph, monitoring (Prometheus, Grafana), Git
**Data & ML:** PySpark, Databricks, Delta Lake, Spark Structured Streaming, ETL pipelines, PostgreSQL, PGVector, embeddings, MLflow, MLOps, LangChain, LLM/RAG, Azure OpenAI, Vertex AI

## Professional Experience

**Advanced Software Engineer** *May 2026 – Present*

*Honeywell | Bengaluru, India*

- Contributed to a performance effort that cut incremental-update install time 32.5% (90m11s to 60m51s on a 44-package run) by merging 13 install sequences into 9, halving a fixed 120s datastore pause and extracting with parallel `pigz`.
- Contributed to EdgeOrchestrator, the Go control-plane service (REST/WebSocket API) that drives Ansible installs and upgrades across 6 deployment profiles and 2 HA topologies on MicroK8s, Ceph and Pacemaker. Playbooks run without a shell, and a marker file stops a crashed self-update reporting success.
- Co-diagnosed a watchdog restart loop wedging idle appliances (Calico IPs left unreleased after hard MicroK8s stops); fixes: registry image pre-caching, a Recreate rollout for the single-writer volume, and a circuit breaker halting auto-restarts after more than 5 in 24 hours.
- Collaborated on ForgeLinkLite, a cross-platform Go agent for Azure DPS enrollment and IoT Hub connectivity (AMQP over WebSocket, certificate auth) with active/passive failover.

*Go, Kubernetes (MicroK8s), Ansible, Ceph, Pacemaker/Corosync, Azure IoT Hub*

**Software Engineer II** *Jul 2024 – Apr 2026*

*Honeywell | Bengaluru, India*

- Built the model deployment service on ForgeML (8 microservices, 10+ tenant workspaces), routing models to Databricks Serving, AKS or edge devices through one REST API with Pydantic discriminated-union validation and two asyncio status monitors.
- Wrote an asyncio Event Hub processor: three parsers behind a Factory Pattern, concurrent PostgreSQL and Unity Catalog writes, and checkpoints committed only after a successful write. Upserts mean a crash replay never duplicates rows.
- Integrated five security scanners (Coverity, Black Duck, Twistlock, ModelScan, AST parser) into model registration behind one Strategy Pattern interface.
- Fine-tuned Gemini on Vertex AI to translate SkySpark Axon rules into Honeywell's Forge DSL, built the versioned prompt store and tuning-job tracking, and validated output against the original rules with domain experts (about 90% correct).
- Shipped the SkySpark migration service (SCRAM auth, Zinc/Trio parsing, parallel writes that roughly halved write time) and a Behave suite of 50+ scenarios run in parallel per tenant, plus Locust load tests.

*Python, FastAPI, Databricks, Azure Kubernetes (AKS), Google Vertex AI, MLflow, Azure Event Hub, PostgreSQL*

**Data Engineer II** *Jun 2023 – Jun 2024*

*Honeywell | Bengaluru, India*

- Contributed to 15+ PySpark ETL pipelines on Databricks ingesting 12+ source systems (SQL Server, Salesforce, SharePoint, field-service databases) into Delta Lake, with timestamp-watermark incremental loads.
- Built a Spark Structured Streaming pipeline over PDFs in Blob Storage: a Delta MERGE per micro-batch for SCD Type 2 history, detection of source deletions, and Change Data Feed so the vectorization job reads only changed rows.
- Contributed to the Maintenance Assist LLM chatbot: agent executor (per-tenant YAML config), async guardrails that cancel unsafe responses mid-stream, a PGVector memory store, and an Azure OpenAI LLM-as-judge evaluation over 1,000+ question-answer pairs.
- Wrote the SQL transformations for field service analytics (resolution trends, contract and maintenance metrics, NPS) that feed the customer-facing DSP dashboard.

*PySpark, Databricks, Delta Lake, PostgreSQL, LangChain, PGVector, Azure OpenAI, MLflow*

## Patent

**Data management in a multi-tenant environment** | US 12,717,808 B2 *Issued Aug 2026*

Co-inventor (3 inventors, Honeywell). Tenant isolation at the data level on a shared multi-tenant backend: records carry tenant metadata and are split and routed by customer, product or a custom rule, with no separate schema or database per tenant.

## Open Source

**DepKeeper** | *Python, Click, httpx, asyncio* *2025 – Present*

Author and maintainer of a CLI that updates `requirements.txt` without crossing a major version, preserving your upper bounds. Dry-run preview, idempotent. On PyPI, Apache 2.0. Merged PRs: aio-libs/async-lru (new `cache_contains()` API) and facebookresearch/exca (five bug fixes).

## Education

**B.Tech Biotechnology + M.Tech Data Science** *2018 – 2023*

*Indian Institute of Technology Madras* *CGPA: 8.77*